Homework #6 — Pre-processing the Adult Dataset

for Predicting Socio-Economic Status (`income > 50K`)

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Deliverables: `6710503739_hw#6_preprocess.ipynb` (analysis) · this report

Dataset. `adult.csv` — **48,842** rows × **15** columns (14 predictors + the `income` label). **Target balance:** 37,155 rows `<=50K` vs 11,687 rows `>50K`, i.e. a **23.93 %** positive rate — the classes are imbalanced at roughly 1:3.

Note on column names. This copy of the dataset spells two columns slightly differently from the attribute description: `educational-num` (description: *education-num*) and `gender` (description: *sex*). The meaning is identical; the file's own spelling is used throughout.

Reproducibility. Every figure and every number in this report was produced by the accompanying notebook, which runs end-to-end without error. The notebook writes its headline statistics to `report_stats.json`, and this report quotes those values, so the two cannot disagree.

1 Phase 1 — Exploratory Data Analysis

1.1 Descriptive statistics for the numerical features

Table 1: Summary statistics for the six numerical features. CV = std / mean, a scale-free measure of spread.

Feature	Mean	Median	Mode	Std	Min	Max	Skew	CV
age	38.64	37.0	36	13.71	17	90	+0.56	0.35
fnlwgt	189,664	178,145	203,488	105,604	12,285	1,490,400	+1.44	0.56
educational-num	10.08	10.0	9	2.57	1	16	−0.32	0.26
capital-gain	1,079.07	0.0	0	7,452.02	0	99,999	**+11.89**	**6.91**
capital-loss	87.50	0.0	0	403.00	0	4,356	**+4.57**	**4.61**
hours-per-week	40.42	40.0	40	12.39	1	99	+0.24	0.31

Interpretation — which features have high variance or significant skew

Variance. In raw units `fnlwgt` dominates everything else (std ≈ 105,604). But `fnlwgt` is a *survey weight* — how many people in the population the row represents — not a property of the person, so its spread is an artefact of the sampling design rather than a signal about income. Comparing raw variances across features with different units is misleading, so the **coefficient of variation** is the fairer measure. On that basis:

- `capital-gain` (CV = 6.91) and `capital-loss` (CV = 4.61) are the genuinely extreme features — their standard deviation is several times their own mean;
- `fnlwgt` is moderate (CV = 0.56);
- `age`, `educational-num` and `hours-per-week` are tight (CV = 0.26–0.35).

Skew. A symmetric feature has skew ≈ 0; |skew| > 1 is normally called strong. Ranked worst first:

- `capital-gain`, skew **+11.89** — by far the worst in the dataset. 91.74 % of rows are exactly 0 and a thin tail runs to 99,999. This is a *zero-inflated* feature, not a bell curve. Its kurtosis of 152.7 confirms an extremely heavy tail.
- `capital-loss`, skew **+4.57** — the same zero-inflated shape (95.33 % zeros).
- `fnlwgt`, skew +1.44 — a moderate right tail.

- `age` (+0.56), `educational-num` (−0.32) and `hours-per-week` (+0.24) are near-symmetric and need no transformation.

The two capital features are the ones that would distort any distance- or gradient-based model, and they are what the outlier strategy in Phase 2 addresses.

1.2 Data-quality audit

Data type of every column

Table 2: Column-by-column type audit: 6 numerical, 9 categorical (one of which is the target).

Column	Kind	Distinct	Notes
age	Numerical (continuous)	74	years, 17–90
workclass	Categorical (nominal)	9	includes the `?` sentinel
fnlwgt	Numerical (continuous)	28,523	survey weight, not a personal attribute
education	Categorical (ordinal)	16	duplicate of `educational-num` (see §1.3.2)
educational-num	Numerical (ordinal)	16	rank code 1–16
marital-status	Categorical (nominal)	7	
occupation	Categorical (nominal)	15	includes the `?` sentinel
relationship	Categorical (nominal)	6	
race	Categorical (nominal)	5	
gender	Categorical (binary)	2	Male / Female
capital-gain	Numerical (continuous)	123	zero-inflated, top-coded at 99,999
capital-loss	Numerical (continuous)	99	zero-inflated
hours-per-week	Numerical (continuous)	96	
native-country	Categorical (nominal)	42	includes the `?` sentinel
income	Categorical (target)	2	`<=50K` / `>50K`

`educational-num` is listed as *ordinal* rather than plain numerical because the distinction matters downstream: an ordinal code must keep its ranking and must *not* be one-hot encoded like a nominal column.

Missing values — the `?` sentinel

`pandas` reports **zero** `NaN` in this file. That is misleading: in the Adult dataset an unknown value is stored as the literal character `?`, which `read_csv` loads as an ordinary string. The audit therefore searched every text column for the sentinel explicitly.

Table 3: Missing values found by searching for the `?` sentinel. All other columns: 0.

Column	# missing	% of rows
occupation	2,809	5.75 %
workclass	2,799	5.73 %
native-country	857	1.75 %
Rows with at least one `?`	**3,620**	**7.41 %**

The missingness is not random. `occupation` and `workclass` are missing on almost exactly the same rows: **2,799 of the 2,809** `occupation` gaps coincide with a missing `workclass`. The 10 rows where `occupation` is missing but `workclass` is present are *all* `Never-worked` — for those people an occupation genuinely does not exist. So "missing" here largely encodes *"not currently in the labour*

force" (retired, unemployed, never worked), which is a structurally meaningful fact rather than a lost measurement. `native-country` is a different case: a small, scattered set of unrecorded values. This distinction drives the imputation decision in §3.1.

Duplicates. 52 rows are exact duplicates of another row (0.11 %).

1.3 Categorical analysis

Distinct labels present in each categorical column

Table 4: Every distinct label found in each categorical column, as it appears in the file.

Column	#	Labels
workclass	9	`?`, Federal-gov, Local-gov, Never-worked, Private, Self-emp-inc, Self-emp-not-inc, State-gov, Without-pay
education	16	10th, 11th, 12th, 1st-4th, 5th-6th, 7th-8th, 9th, Assoc-acdm, Assoc-voc, Bachelors, Doctorate, HS-grad, Masters, Preschool, Prof-school, Some-college
marital-status	7	Divorced, Married-AF-spouse, Married-civ-spouse, Married-spouse-absent, Never-married, Separated, Widowed
occupation	15	`?`, Adm-clerical, Armed-Forces, Craft-repair, Exec-managerial, Farming-fishing, Handlers-cleaners, Machine-op-inspct, Other-service, Priv-house-serv, Prof-specialty, Protective-serv, Sales, Tech-support, Transport-moving
relationship	6	Husband, Not-in-family, Other-relative, Own-child, Unmarried, Wife
race	5	Amer-Indian-Eskimo, Asian-Pac-Islander, Black, Other, White
gender	2	Female, Male
native-country	42	`?`, Cambodia, Canada, China, Columbia, Cuba, Dominican-Republic, Ecuador, El-Salvador, England, France, Germany, Greece, Guatemala, Haiti, Holand-Netherlands, Honduras, Hong, Hungary, India, Iran, Ireland, Italy, Jamaica, Japan, Laos, Mexico, Nicaragua, Outlying-US(Guam-USVI-etc), Peru, Philippines, Poland, Portugal, Puerto-Rico, Scotland, South, Taiwan, Thailand, Trinadad&Tobago, United-States, Vietnam, Yugoslavia
income (target)	2	`<=50K`, `>50K`

Note that `?` appears in the label list of exactly the three columns identified above — confirming the sentinel is being read as a legitimate category and must be handled before encoding. `native-country` is heavily concentrated: **91.4 %** of the known values are `United-States`, and most of the remaining 40 countries hold well under 1 % each.

`education` vs `educational-num` — why keeping both is redundant

Grouping the data by `education` shows that each of the 16 text labels maps to **exactly one** `educational-num` value, and each of the 16 numeric values maps back to exactly one label — a verified **perfect one-to-one mapping**:

`Preschool`→1, `1st-4th`→2, `5th-6th`→3, `7th-8th`→4, `9th`→5, `10th`→6, `11th`→7, `12th`→8, `HS-grad`→9, `Some-college`→10, `Assoc-voc`→11, `Assoc-acdm`→12, `Bachelors`→13, `Masters`→14, `Prof-school`→15, `Doctorate`→16

The two columns therefore carry *exactly* the same information — one variable expressed twice, once as text and once as an integer code. Keeping both is redundant and actively harmful:

1. **Perfect multicollinearity.** A model given both sees two features related deterministically. For linear models the coefficients become unidentifiable (the design matrix is rank-deficient); for tree ensembles the importance of "education" is split arbitrarily between two identical columns, so neither looks important.
2. **Wasted dimensionality.** One-hot encoding `education` would add 16 sparse columns that duplicate a single existing integer column.
3. **Loss of order.** `educational-num` already encodes the *rank* (a Doctorate outranks a Bachelors). One-hot encoding the text version discards that ordering entirely.

Decision: drop `education`, keep `educational-num`, which is already a correct ordinal encoding. §3.2 proves this by re-deriving the ordinal code from scratch with an explicitly stated rank order and confirming it reproduces `educational-num` value-for-value.

2 Phase 2 — Visual Discovery & Outlier Handling

2.1 Distributions of `age`, `hours-per-week` and `capital-gain`

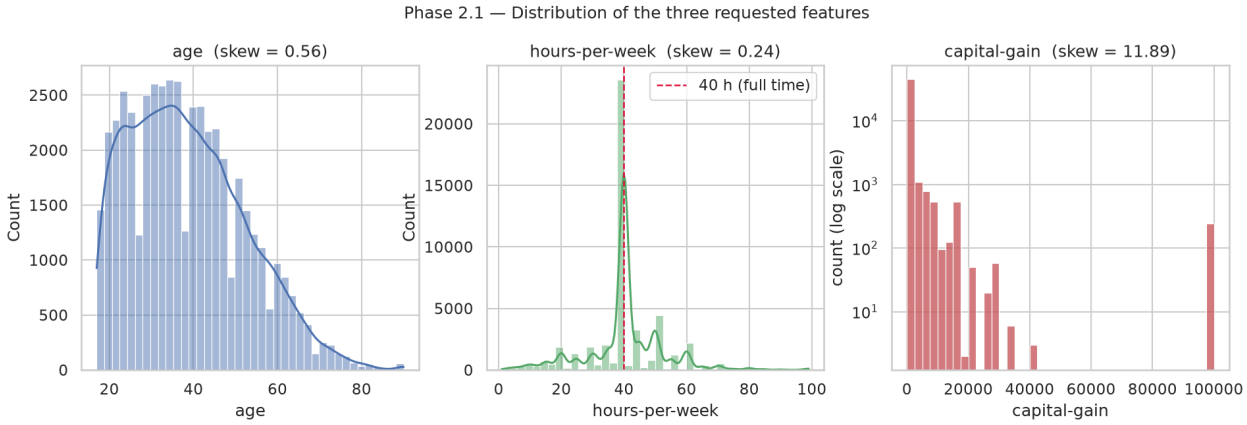


Figure 1: Histograms of the three requested features. `capital-gain` uses a logarithmic count axis — on a linear axis only the zero bar is visible.

- **`age`** — right-skewed but entirely plausible: a peak in the late twenties to thirties, a long thin tail into old age, hard-stopped at exactly 90.
- **`hours-per-week`** — a huge spike at exactly 40 (the standard working week), with smaller spikes at 20, 45, 50 and 60. This is not a smooth distribution but a mixture of socially conventional values, which is why its interquartile range is only 5 hours.
- **`capital-gain`** — visually a single bar at zero. On the log-count axis the tail appears, and a **separate isolated bar sits exactly at 99,999**, clearly detached from the rest of the distribution.

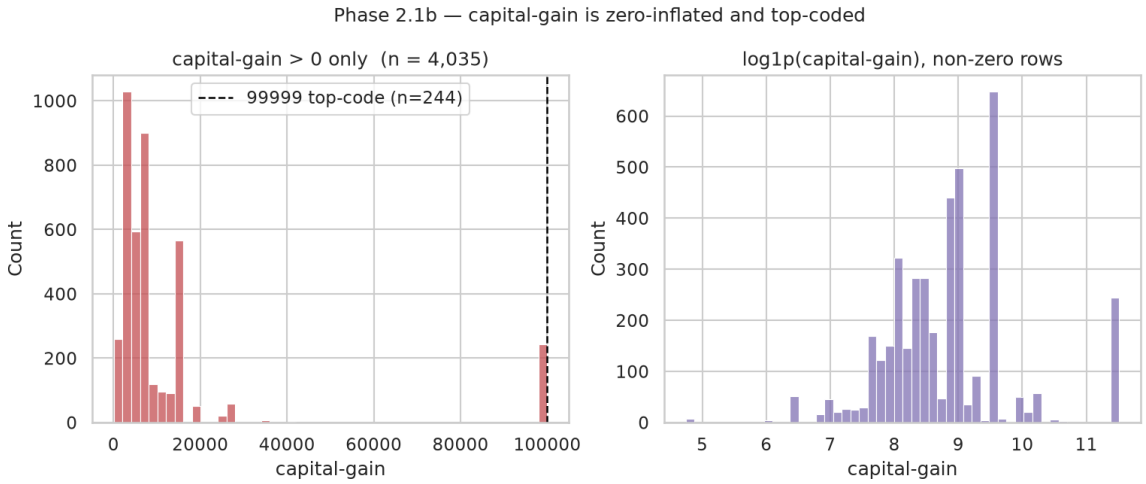


Figure 2: `capital-gain` restricted to its 4,035 non-zero rows. Left: the raw values, with the 99,999 ceiling marked. Right: the same values after `log1p`.

Zooming in confirms it: the largest value *below* the ceiling is **41,310**, and then **244 rows** sit on exactly **99,999** with nothing in between. No natural distribution produces a cluster on a single round ceiling — this is census **top-coding**: every gain above the cut-off was clipped to 99,999.

2.2 Box plots and IQR outlier counts

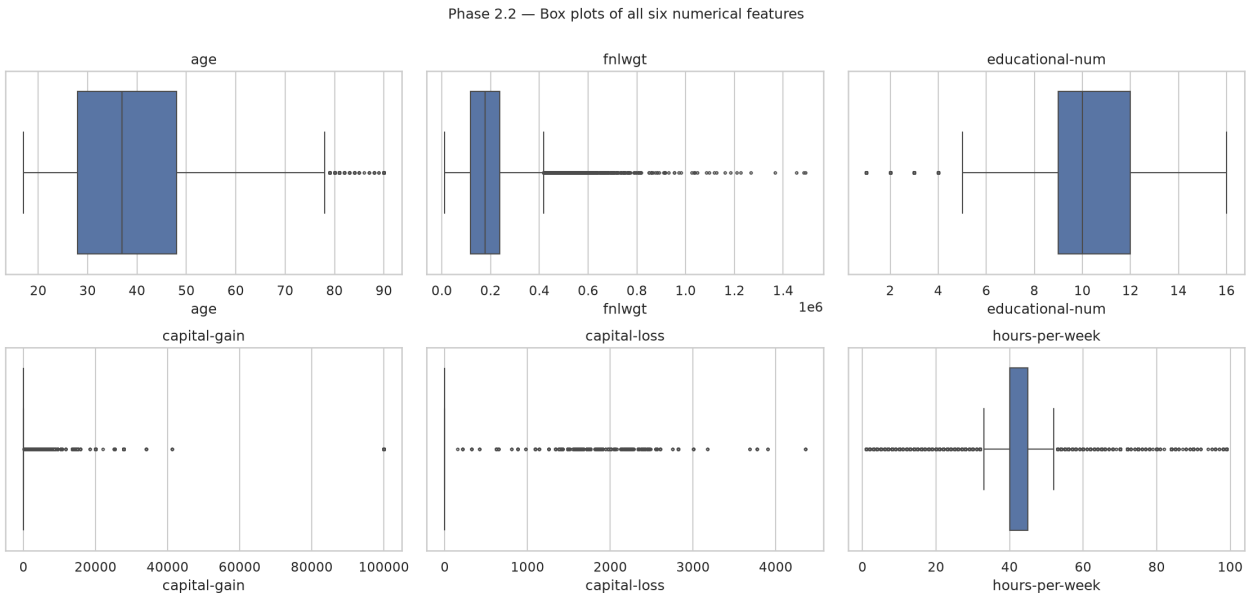


Figure 3: Box plots of all six numerical features on their raw scales.

Table 5: Outliers by the standard 1.5×IQR rule. Note that for `capital-gain` and `capital-loss` the IQR is *zero* (Q1 = Q3 = 0), so the rule flags *every* non-zero value as an outlier.

Feature	Q1	Q3	IQR	Lower fence	Upper fence	# (%) flagged
age	28.0	48.0	20.0	−2.0	78.0	216 (0.44 %)
fnlwgt	117,551	237,642	120,092	−62,587	417,779	1,453 (2.97 %)
educational-num	9.0	12.0	3.0	4.5	16.5	1,794 (3.67 %)
capital-gain	0.0	0.0	0.0	0.0	0.0	4,035 (8.26 %)
capital-loss	0.0	0.0	0.0	0.0	0.0	2,282 (4.67 %)
hours-per-week	40.0	45.0	5.0	32.5	52.5	13,496 (27.63 %)

2.3 Outlier strategy

Feature chosen: `capital-gain` — the most extreme feature in the dataset (skew +11.89, maximum 99,999, every non-zero value flagged by the IQR rule). `age` is discussed as a secondary case.

Are these errors or real-world extremes?

Both — in different ways, and the evidence separates them cleanly:

Observation	Reading
91.74 % of rows are exactly 0	**Real.** Most people realise no capital gain in a year. Zero is a true value, not a gap — so this is not a missing-data problem.
A smooth tail from a few hundred up to 41,310	**Real-world extremes.** Genuine investment income. These people really do earn more, and the values behave like a continuous distribution.
Exactly 244 rows at exactly 99,999, with a gap of ≈ 58,000 below them	**An artefact.** Census **top-coding** — everything above the cut-off was clipped to one value. 99,999 is a censoring flag, not a measurement.

Before deciding, the notebook checked whether the tail is actually informative:

Table 6: `>50K` rate by `capital-gain` band.

capital-gain band	# rows	% earning `>50K`
0	44,807	20.5 %
1–5,000	1,584	17.9 %
5,000–20,000	2,070	88.7 %
20,000–99,998	137	92.7 %
99,999 (top-coded)	244	**100.0 %**

The relationship is *not* monotonic at the bottom — the rate dips slightly from 20.5 % at zero to 17.9 % for small gains (small gains come mostly from modest earners) — but above 5,000 it jumps to 88.7 %, then 92.7 %, and reaches **100 %** for the top-coded group. The tail is therefore among the strongest predictors in the dataset, and the top-coded rows are not junk to be discarded: they are the richest people in the sample.

Decision: KEEP the outliers, but transform them

Option	Verdict
Remove the rows	× **Rejected.** The 1.5×IQR rule flags 8.26 % of rows on `capital-gain` alone, and **41.53 % of the entire dataset (20,284 rows)** once every numerical column is considered — `hours-per-week` alone flags 27.63 %, because its IQR is a mere 5 hours. Worse, removal would delete almost every high earner, i.e. exactly the signal the model is being built to detect, biasing the target distribution.
Replace (winsorise, or impute with the median)	× **Rejected as the primary treatment.** These are not measurement errors. Replacing 99,999 with the median would destroy the single strongest predictor in the dataset (a group with a 100 % `>50K` rate).
Keep, but reshape the scale	✓ **Chosen.** Retains every row and every ordering, while removing the numerical dominance that the raw scale would give the feature.

Concretely, this decision is implemented in Phases 3–4 as:

1. **Keep every row.** No row is deleted for being an outlier. (The 52 exact duplicates *are* dropped — but as a data-collection artefact, not as outliers.)
2. **Apply `log1p`** to `capital-gain` and `capital-loss`. This compresses the skew towards symmetry so the feature can no longer dominate a distance metric, while preserving the *order* of every value. `log1p` rather than `log` because it maps the 91.74 % of exact zeros to 0 instead of $-\infty$.
3. **Add an explicit `capital-gain-topcoded` flag** for the 244 censored rows, so a model can learn "this value is a ceiling, not a measurement" rather than treating 99,999 as a precise amount.

`age` is treated differently and more simply. The IQR rule flags 216 people over 78, but a 90-year-old is a perfectly real survey respondent, so those rows are **kept untransformed**. The pile-up at exactly 90 is the same top-coding artefact (age was capped at 90), but with a negligible effect on the feature's scale, so no transform is warranted.

3 Phase 3 — Data Pre-processing & Transformation

3.1 Handling missing values

Step 1 — make the missingness visible. All text columns are whitespace-stripped and the `?` sentinel is replaced with a real `NaN`, so that standard pandas missing-value tooling works at all.

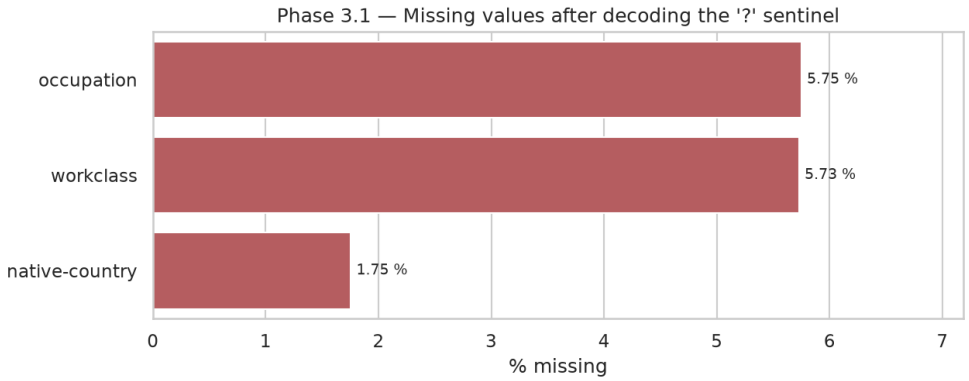


Figure 4: Missing values once the `?` sentinel has been decoded to `NaN`.

Step 2 — choose the strategy. The audit in §1.2 showed these gaps are not one homogeneous problem, so a single blanket rule would be wrong. **Two different strategies** are applied.

(a) `workclass` and `occupation` → a new explicit category `"Unknown"`

Why. These two columns are missing on the same 2,799 rows, and the pattern is structural: these are people outside the labour force. "No workclass" is a *fact about the person*, not a lost value. The two groups are demonstrably different populations:

	# rows	Mean age	Mean hours/week	% `>50K`
`workclass` missing	2,795	40.2	31.8	**9.5 %**
`workclass` present	45,995	38.6	41.0	**24.8 %**

The missing group is older, works roughly nine fewer hours a week, and is about **2.6 times less likely** to earn `>50K`. Imputing with the mode (`Private` / `Prof-specialty`) would *fabricate a job for 2,795 retired or unemployed people* and erase a genuinely predictive signal. Encoding the gap as its own `"Unknown"` category keeps that signal available to the model and — since these columns are about to be one-hot encoded anyway — costs only one extra binary column each.

(b) `native-country` → mode imputation (`United-States`)

Why. Only 1.75 % of rows are affected, there is no structural story behind the gap (an unrecorded country is simply an unrecorded field, not a statement about the person), and the column is overwhelmingly dominated by one value — **91.4 %** of all *known* countries are `United-States`. Mode imputation on a 1.75 % gap in a 91 %-dominant column shifts the column's distribution by well under two percentage points, and avoids inflating the one-hot matrix with a 42nd category that would carry almost no information.

After imputation the dataset contains **zero** missing values (asserted in the notebook).

3.2 Dropping the redundant column — with proof

Before dropping `education`, the ordinal encoding was rebuilt from scratch using scikit-learn's `OrdinalEncoder` with the academic rank order stated explicitly (`Preschool` < `1st-4th` < ⋯ < `Prof-school` < `Doctorate`). The hand-built code reproduces `educational-num` exactly — **value for value, across all 48,790 rows** — which confirms the redundancy claim of §1.3.2 empirically rather than by assertion. `education` is then dropped.

3.3 Categorical encoding

Three different encodings are used, because these are three different kinds of variable. Using one-hot everywhere would throw away a real ordering; using label encoding everywhere would invent an ordering that does not exist.

Kind	Columns	Encoding	Reason
Ordinal	`educational-num`	already a 1–16 rank code	The ranking is real and meaningful (a Doctorate outranks a Bachelors), so preserve it.
Binary	`gender`, `income` (target)	label encode 0/1	With two levels, one-hot and label encoding are mathematically equivalent; 0/1 is simpler and adds no columns. `Male`=1, `>50K`=1.
Nominal	`workclass`, `marital-status`, `occupation`, `relationship`, `race`, `native-country`	one-hot, `drop_first=True`	There is no sensible ordering between e.g. `Sales` and `Tech-support`; a label code would make a model believe one is "greater than" the other.

An extra engineered column, `capital-gain-topcoded`, is added here per the Phase 2 decision.

Why `drop_first=True`. With k levels only $k-1$ indicator columns are needed — the dropped reference level is implied when all the others are 0. Keeping all k creates the *dummy-variable trap*: the columns sum to 1 for every row, so the design matrix is rank-deficient and linear-model coefficients become unidentifiable. This also removes 6 redundant columns.

Result: 15 columns → 86 columns (48,790 rows), all numeric.

4 Phase 4 — Scaling & Final Preparation

4.1 The skew transform (applied before scaling)

Scaling does *not* fix skew — standardising `capital-gain` would leave a feature with skew 11.89, merely re-centred. The `log1p` transform decided in §2.3 is therefore applied first:

Feature	Skew before `log1p`	Skew after `log1p`
capital-gain	+11.89	**+3.11**
capital-loss	+4.57	+4.30

Reported honestly: `capital-gain` improves dramatically — skew 11.89 → 3.11, and its range collapses from 0–99,999 to 0–11.5, so it can no longer dominate a distance metric. `capital-loss` barely moves (4.57 → 4.30), and that is expected: 95.33 % of its values are exactly 0, so the residual skew comes from the *zero-inflation itself*, not from the scale of the non-zero tail — and no monotone transform can remove a point mass at zero. The residual skew is accepted here; if a later model proved sensitive to it, the next step would be to split the column into a binary `has-capital-loss` indicator plus the log amount.

4.2 Train/test split *before* fitting the scaler

The data is split 80/20, stratified on `income`, *before* the scaler is fitted: **39,032 training** rows and **9,758 test** rows, each retaining the 23.9 % `>50K` rate. The scaler is then fitted on the training set only and merely *applied* to the test set. Fitting it on the full dataset would leak the test set's mean and standard deviation into training and produce an optimistically biased estimate of model performance later.

4.3 Feature scaling

Standardisation (`StandardScaler`, $z = (x-\mu)/\sigma$) is chosen as the primary method:

- it does not depend on the minimum and maximum, so a single extreme value cannot squash every other observation into a narrow band — which matters here precisely because Phase 2 decided to *keep* the extremes;
- it is the assumption behind regularised linear models, SVMs and PCA;
- the 79 binary columns (77 one-hot indicators, plus `gender` and `capital-gain-topcoded`) are deliberately **left untouched** — they are already on a 0/1 scale, and standardising them would destroy their sparsity and their interpretability. Only the six genuinely continuous columns are scaled.

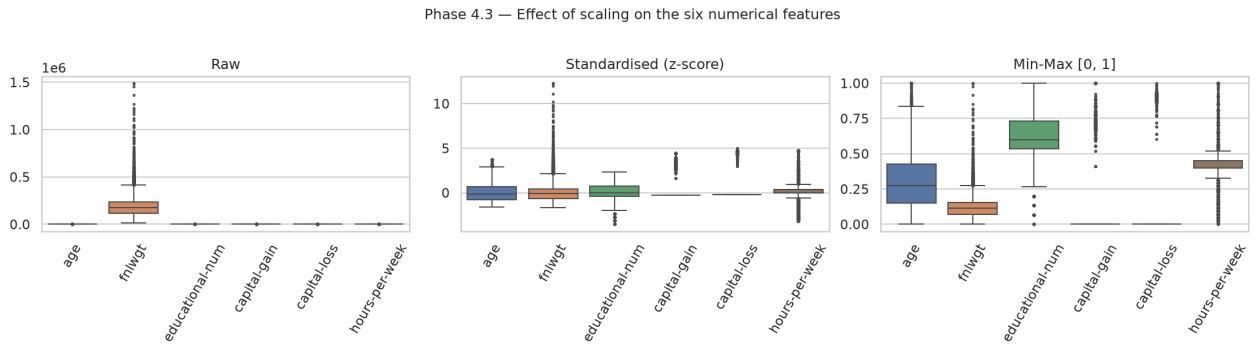


Figure 5: The six numerical features before scaling, after standardisation, and after Min-Max scaling (training set).

The left panel shows the problem: on the raw axis `fnlwgt` (tens of thousands) is the only feature visible at all — `age`, `educational-num` and `hours-per-week` are flattened to a line near zero. A distance-based model would be deciding almost entirely on `fnlwgt`, which as noted in §1.1 is a survey weight carrying no information about the person.

Both scalers fix that, but the instructive difference is **`fnlwgt`**: its maximum (1,490,400) is roughly six times its own Q3, so Min-Max — which anchors on that single most extreme value — squeezes its entire interquartile range into about the 0.07–0.15 band, whereas standardisation leaves a box of normal size. That sensitivity to one extreme value is exactly why standardisation was selected for the final output.

(`capital-gain` and `capital-loss` appear as flat lines in *both* scaled panels. That is not a scaling failure but the zero-inflation of §2.3 showing through: with 92 %/95 % of values identical, Q1 = median = Q3, so the box has no width at any scale.)

4.4 Final verification and output

All six automated checks in the notebook pass:

✓ no missing values remain	✓ target is strictly 0/1
✓ all 86 columns are numeric	✓ scaled means = 0 (to 10^{-9})
✓ no duplicate rows	✓ scaled standard deviations = 1 (to 10^{-9})

Files written by the notebook: `adult_train_preprocessed.csv` (39,032 rows), `adult_test_preprocessed.csv` (9,758 rows), and `adult_encoded_unscaled.csv` (48,790 rows, encoded but unscaled, for models such as tree ensembles that do not need scaling).

5 Summary of the pipeline

#	Step	Decision	Why
1	Load	48,842 × 15	—
2	Duplicates	dropped 52 exact duplicates	identical records are a collection artefact, not information
3	Missing `?`	decoded to `NaN`	pandas reports 0 `NaN` otherwise
4	`workclass`, `occupation`	→ `"Unknown"` category	the gap is structural (outside the labour force) and is itself predictive
5	`native-country`	→ mode `United-States`	1.75 % gap, no structural story, 91.4 %-dominant column
6	`education`	dropped	verified one-to-one duplicate of `educational-num`
7	Outliers	**kept**, not removed	real-world extremes carrying the strongest income signal; removal would cost 41.5 % of rows
8	`capital-gain` = 99999	flagged with a new binary column	top-coding artefact, not a measurement
9	`capital-gain/loss`	`log1p`	skew 11.89 → 3.11 while preserving order (`capital-loss` stays skewed — zero-inflation, not scale)
10	`gender`, `income`	label encode 0/1	binary
11	6 nominal columns	one-hot, `drop_first=True`	no natural order; avoids the dummy-variable trap
12	Split	80/20 stratified, *before* scaling	prevents test-set leakage into the scaler
13	6 continuous columns	`StandardScaler`	robust to the retained extremes; dummies left as 0/1

Outcome. The raw 48,842 × 15 table with a hidden `?` sentinel, a duplicated column, an 11.9-skew feature and nine text columns has become a **48,790 × 86 fully numerical, gap-free, comparably scaled matrix**, split into stratified train and test sets and ready for a classification model.